

Classification of Human Activities Exploration of Preprocessing Pipeline Meta 1 Project

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October 2025

1 Introduction

Human Activity Recognition (HAR) is a rapidly evolving field with applications in healthcare, fitness, ambient assisted living, and human-computer interaction. The classification of physical activities—such as walking, sitting, or climbing stairs—based on sensor data is a central task in HAR. The core idea of this study is to explore a possible pipeline approach of activity classification, from data preprocessing and outlier detection to feature extraction and selection, using the FORTH-TRACE benchmark dataset. The main idea is to develop a robust preprocessing pipeline for the possible future use of machine learning techniques.

2 Related Work

There are several works in this area and in this dataset. The statistical foundations for evaluating and comparing activity data are critical to this task. Du Prel et al. (2010) [1] provide a comprehensive guide to selecting appropriate statistical tests based on data type, distribution, and study design. Their framework emphasizes the importance of distinguishing between paired and unpaired samples, and between parametric and non-parametric methods—principles that directly inform the analysis of sensor-derived features in HAR. Additionally, the benchmark study by Karagiannaki et al.

(2016) [2] highlights the impact of feature selection on classification performance, comparing various selection strategies across HAR datasets. These works collectively underscore the methodological rigor required for effective activity classification and guide the statistical and computational choices throughout this assignment.

3 Materials

3.1 Data Description

The dataset used in this work is the FORTH-TRACE benchmark¹, designed for human activity recognition using multimodal wearable sensors. It includes data from 15 participants, each equipped with five body-worn devices positioned on the wrists, chest, and legs. Each participant's data is distributed across five CSV files - one per device - containing readings from accelerometers, gyroscopes, and magnetometers along three axes, as well as computed magnitudes and activity labels. After vertical merging of all files (due to consistent column structure), the final dataset comprises 3 930 873 rows and 13 columns. This structure enables comprehensive temporal and spatial analysis of human motion across 16 distinct activities.

The features, it's description and the activities description are in the figures below.

3.2 Tools and Packages

The project was implemented using the following tools and packages:

- **Pandas, Numpy:** for data manipulation tasks.
- **Scipy:** Utilized for statistics used in feature significance.
- **Scikit-Learn:** Utilized to build custom outlier detection models and for feature engineering tasks.
- **Matplotlib, Seaborn:** Used to visualize data and understand trends.

¹**FORTH-TRACE Benchmark Dataset [13]:** https://github.com/spl-icsforth/FORTH_TRACE_DATASET/

Feature	Description
Participant	Identifier for the participant
Device ID	Identifier for the device (redundant)
accelerometer x	Acceleration along the X-axis
accelerometer y	Acceleration along the Y-axis
accelerometer z	Acceleration along the Z-axis
gyroscope x	Angular velocity along the X-axis
gyroscope y	Angular velocity along the Y-axis
gyroscope z	Angular velocity along the Z-axis
magnetometer x	Magnetic field strength along the X-axis
magnetometer y	Magnetic field strength along the Y-axis
magnetometer z	Magnetic field strength along the Z-axis
Timestamp	Time of data acquisition
Activity Label	Numeric code representing the performed activity

Table 1: Dataset features and their descriptions after importation.

Label	Activity Description
1	Stand
2	Sit
3	Sit and Talk
4	Walk
5	Walk and Talk
6	Climb Stairs (up/down)
7	Climb Stairs (up/down) and Talk
8	Transition: Stand $\rightarrow$ Sit
9	Transition: Sit $\rightarrow$ Stand
10	Transition: Stand $\rightarrow$ Sit and Talk
11	Transition: Sit $\rightarrow$ Stand and Talk
12	Transition: Stand $\rightarrow$ Walk
13	Transition: Walk $\rightarrow$ Stand
14	Transition: Stand $\rightarrow$ Climb Stairs (w/wo talking)
15	Transition: Climb Stairs $\rightarrow$ Walk
16	Transition: Climb Stairs and Talk $\rightarrow$ Walk and Talk

Table 2: Activity Labels and Descriptions

4 Exploratory Data Analysis for Preprocessing

4.1 Outlier Detection

Outlier detection is a critical step in data preprocessing, particularly in sensor-based activity recognition tasks. Outliers - data points that deviate significantly from expected patterns - can arise due to sensor noise, recording errors, or atypical user behavior. If left untreated, these anomalies can distort statistical analyses, bias feature extraction, and degrade model performance. Identifying and handling outliers ensures cleaner input data, improves the reliability of descriptive statistics, and enhances the robustness of machine learning models.

For this task, the outliers of the dataset were analysed with 3 different methods: IQR, Z-score and K-means.

4.1.1 IQR Method

Outlier detection was performed using the Interquartile Range (IQR) method, following Tukey’s definition as implemented in boxplot visualizations. The IQR is defined as the range between the first (Q1) and third (Q3) quartiles of a distribution. Data points falling below $Q1 - 1.5 \cdot IQR$ or above $Q3 + 1.5 \cdot IQR$ are considered outliers. Also, this approach is robust to non-normal distributions.

Visual inspection of the boxplots (figures 1, 2 and 3) and the outlier density in figure 11 revealed consistent patterns across the transformed dataset, which includes the magnitudes of acceleration, gyroscope, and magnetometer vectors. Activities involving dynamic motion - such as walking, climbing stairs, and transitions (especially while talking) - exhibited higher variability and a greater spread of outliers. In contrast, static activities like sitting and standing showed tighter distributions with fewer extreme values. These trends suggest that movement complexity and a possible sensor instability contribute to outlier density.

4.1.2 Z-score Method

To complement the IQR-based analysis, outlier detection was also performed using the Z-score method. This approach identifies outliers by measuring how many standard deviations a data point deviates from the mean. For a given threshold k , any sample with a Z-score greater than k or less than $-k$ is considered an outlier. In addition, this method is sensitive to extreme values in the distribution.

Outliers were detected across the magnitudes of acceleration, gyroscope, and magnetometer vectors using thresholds $k = 3.0, 3.5$, and 4.0 . Visualizations were generated per activity, highlighting outliers in red and regular samples in blue, as we can visualize in the figures 4, 5 and 6. As expected, increasing the threshold k resulted in a linear decrease in the number of detected outliers, confirming the method’s sensitivity to the chosen cutoff.

A comparative analysis with the IQR method revealed similar trends, especially in the accelerometer data. Activities involving dynamic movement showed higher outlier densities, while static activities remained more stable.

4.1.3 K-Means Method

Outlier detection was further explored using the K-means clustering algorithm applied to the 3D space formed by the magnitudes of the module of acceleration, gyroscope, and magnetometer vectors (figures 7,8,9 and 10). This representation captures the overall intensity of motion and sensor activity, offering a compact and interpretable feature space for clustering. This time, K-Means was only applied to device 2 in order to make comparisons with the other methods

We applied K-means with `n_clusters = {3, 4, 5}` to this space to capture increasing granularity in behavioral patterns. For each configuration, the Euclidean distance of each sample to its assigned cluster centroid was computed. Outliers were initially identified using a global threshold based on the 95th percentile of these distances. This approach consistently flagged approximately 5% of samples as outliers per activity, reflecting the fixed percentile cutoff. We can see the outliers per k value in figure 7.

However, this uniform rate raised concerns about overgeneralization. To refine the detection, we implemented a second approach using a dynamic threshold defined as:

$$\text{threshold} = \mu + k \cdot \sigma$$

where μ is the mean distance to the centroid and σ is the standard deviation. This method allowed for activity-dependent sensitivity, producing outlier densities more aligned with those observed in the Z-score and IQR analysis.

4.2 Comparison of Outlier Detection Methods

4.2.1 IQR vs. Z-Score

Visual and quantitative analysis revealed consistent trends between the two methods. Activities involving dynamic transitions - such as walking, climbing stairs, and posture changes - exhibited higher outlier densities due to abrupt sensor fluctuations. Static activities like sitting and standing showed lower variability and fewer anomalies. The Z-score method demonstrated a linear sensitivity to the threshold k , with higher values yielding fewer outliers.

4.2.2 K-Means vs. Univariate Methods

Initially, outliers were identified using a fixed global threshold based on the 95th percentile of cluster distances. This approach consistently flagged approximately 5% of samples per activity, regardless of distribution. However, this uniformity limited comparability with IQR and Z-score, which showed activity-dependent densities. This adjustment allowed K-Means to reflect true distributional extremes and align more closely with the patterns observed in the univariate methods.

4.2.3 Summary of Observations

Observations from all figures, including the outlier density charts, we took this summary conclusions:

- **IQR and Z-score** revealed variable outlier densities across activities, with higher rates in transitional and complex movements.
- **K-Means (adjusted)** produced similar density patterns, particularly effective in identifying multivariate anomalies.
- **Percentile-based thresholds** in K-Means are simple to implement but can artificially constrain outlier rates.
- **Activity-dependent thresholds** are essential for meaningful comparisons and accurate anomaly detection.

5 Feature Extraction

5.1 Significance Analysis

To evaluate the discriminative power of the selected variables across different activities, we performed a statistical significance analysis of their mean values. We investigate the methods in [1] for this purpose. First, we assessed the normality of each variable’s distribution using the Kolmogorov-Smirnov test. The results indicated that none of the variables met the normality assumption ($p < 0.05$), justifying the use of non-parametric methods.

Consequently, we applied the Kruskal-Wallis test to compare the distributions of each variable across activity classes. This test revealed statistically significant differences for all variables ($p < 0.05$), confirming that sensor readings vary meaningfully between activities.

The analysis included individual components of the accelerometer, gyroscope, and magnetometer across all axes (x , y , z), as well as their respective magnitude vectors. All variables demonstrated significant variation, suggesting that no single axis or sensor dominates in discriminative power. Instead, each contributes meaningfully to the classification of human activities. This conclusions can be verified in table 3.

5.2 Temporal and Spectral Feature Extraction

Following the methodology outlined in references [2], [3], and [4], we developed a comprehensive feature extraction pipeline based on sliding windows of 5 seconds with 50% overlap. Given the average sampling interval of 20.47 ms, the effective sampling frequency is approximately 48.86 Hz, resulting in roughly 2443 samples per window. Any window spanning multiple activity labels was discarded to preserve class consistency.

The extracted feature set comprises 178 features derived from the accelerometer, gyroscope, and magnetometer signals across all three axes (x , y , z). These features fall into three main categories:

5.2.1 Statistical Features (108)

For each of the 9 sensor axes (ax , ay , az , gx , gy , gz , mx , my , mz), we computed 12 statistical descriptors:

- Mean: Average signal value
- Median: Median signal value

- Standard Deviation: Spread of the signal
- Variance: Square of the standard deviation
- Root Mean Square (RMS): Quadratic mean
- Averaged Derivatives: Mean of first-order differences
- Skewness: Asymmetry of the distribution
- Kurtosis: Peakedness of the distribution
- Interquartile Range (IQR): Difference between 75th and 25th percentiles
- Zero Crossing Rate (ZCR): Rate of sign changes
- Mean Crossing Rate (MCR): Rate of crossings over the mean
- Spectral Entropy: Frequency distribution complexity

5.2.2 Correlation Features (36)

We computed pairwise Pearson correlations between all combinations of the 9 axes, resulting in 36 correlation features. These capture inter-axis relationships and signal synchrony.

5.2.3 Physical and Spectral Features (34)

Based on the framework proposed in [3], we extracted 34 physics-informed and spectral features:

- Movement Intensity (MI), Signal Magnitude Area (SMA)
- Eigenvalues of Dominant Directions (EVA1, EVA2)
- Correlation between Acceleration and Heading (CAGH)
- Averaged Velocity along Heading (AVH) and Gravity (AVG)
- Averaged Rotation Angles related to Gravity (ARATG)
- Dominant Frequency (DF) and Energy (EN) per axis
- Aggregated Acceleration Energy (AAE), Rotation Energy (ARE), and Magnetometer Energy (AME)
- Binned Velocity along Heading (AVH_bin1 to AVH_bin5)

5.2.4 Feature Count Justification

The total number of features is computed as follows. Also, check the table 4:

- 12 statistical features $\times$ 9 axes = 108
- 36 pairwise correlation features
- 34 physical and spectral features

$$\text{Total Features} = 108 + 36 + 34 = 178$$

6 Feature Selection

To identify the most discriminative features for activity classification, we applied three feature selection methods: Principal Component Analysis (PCA) as an unsupervised approach, and Fisher Score and ReliefF as supervised methods.

6.1 PCA Application

We applied PCA to the normalized feature set using Z-score standardization. The goal was to reduce dimensionality while retaining most of the variance. The analysis of figure 14 showed that 37 principal components (PCs) were sufficient to explain 75% of the total variance.

Each sample was projected onto these 37 components, forming a compressed representation. For example, a feature vector $\mathbf{x}_t$ is transformed into $\mathbf{z}_t = \mathbf{W}_{37}^\top \cdot \mathbf{x}_t$, where $\mathbf{W}_{37}$ contains the top components.

Advantages:

- Reduces computational cost
- Removes feature redundancy
- Facilitates visualization and clustering
- Can improve supervised model performance

Limitations:

- Loss of interpretability (components lack physical meaning)

- May discard variance relevant for classification
- Assumes linearity (misses non-linear relationships)
- Sensitive to feature scaling

6.2 Fisher Score Algorithm

This algorithm evaluates each feature by comparing between-class variance to within-class variance. Features with high Fisher Scores exhibit strong class separation.

6.3 ReliefF Algorithm

This method estimates feature relevance by measuring how well feature values distinguish between neighboring samples of different classes. It captures local dependencies and interactions, including linear and non-linear relationships.

6.3.1 Top Features and Comparison

Using both methods, we selected the top 10 features as shown in tables 5 and 6. Notably, `EN_mx` and `mz_median` appeared in both rankings, indicating consistent importance across selection strategies.

6.3.2 Feature Retrieval

To obtain the selected features for a given sample $\mathbf{x}_t$, we index the original feature vector using the ranked feature indices. For example, if the top 10 features are stored in a list `selected_features`, then:

$$\mathbf{x}_t^{\text{selected}} = \text{selected_features}(\mathbf{x}_t)$$

This yields a reduced feature vector containing only the most relevant dimensions.

Advantages and Limitations of these methods

Advantages:

- Improves model performance by focusing on informative features
- Reduces dimensionality and computational cost
- Enhances interpretability by highlighting key variables

Limitations:

- May overlook feature interactions (especially Fisher Score)
- Sensitive to noise and class imbalance
- ReliefF requires careful tuning of neighbors and may be computationally intensive

6.4 Conclusion

The combined approach of outlier detection and feature engineering proved essential for preparing high-quality sensor data for activity recognition. Outlier detection methods - namely IQR, Z-score, and K-Means - helped identify and filter anomalous samples, improving data reliability and reducing noise. Feature extraction, both temporal and spectral, enriched the dataset with meaningful descriptors that capture statistical, physical, and frequency-based patterns. Together, these steps laid a robust foundation for effective feature selection and model training, ensuring that the input data is both clean and informative.

7 Acknowledgments

This project was a key learning experience in understanding Feature Engineering. From data preprocessing to feature selection, every step helped us handle the challenges of building better pipelines for further machine learning analysis.

Also, we would like to mention that AI [ChatGPT] was used to:

1. Correct some syntax errors and make the text more cohesive and coherent.
2. Correct chart plotting code.
3. creating the feature extraction functions in order to obtain the 178 new features.
4. Exploring feature selection models

8 References

a) Articles and Papers

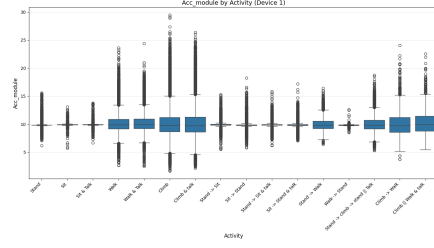
1. 3 Jean-Baptist du Prel, Bernd Röhrig, Gerhard Hommel, Maria Bletner, Choosing Statistical Tests, Deutsches Arzteblatt, v107(19), 2010. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2881615/>
2. Katerina Karagiannaki, Athanasia Panousopoulou, Panagiotis Tsakalides, A Benchmark Study on Feature Selection for Human Activity Recognition, UBICOMP/ISWC '16,,<https://dl.acm.org/doi/pdf/10.1145/2968219.2971421>
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4. Enhanced Human Activity Recognition (HAR): Leveraging Sub-Window Techniques and Feature Ratios from Triaxial Accelerometer Data. (2024). International Journal of Intelligent Engineering and Systems, 18(1), 481–498. <https://doi.org/10.22266/ijies2025.0229.34>

b) Code References

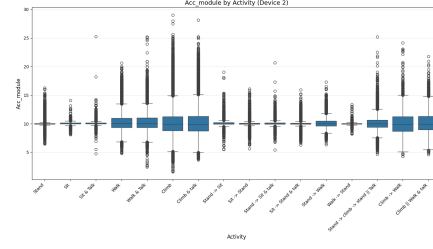
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11. <https://medium.com/@yashdagli98/feature-selection-using-relief-algorithms-with-python-example-3c2006e18f83>
12. https://scikit-learn.org/stable/modules/feature_selection.html
- c) Dataset
13. https://github.com/spl-icsforth/FORTH_TRACE_DATASET/

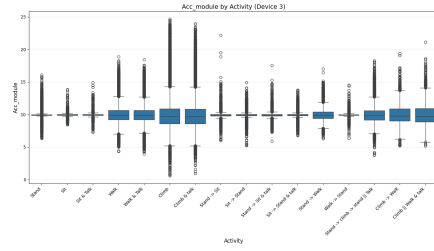
9 Figure and Tables



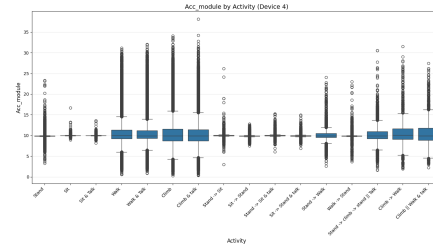
(a) Boxplots of accelerometer by each activity for device 1.



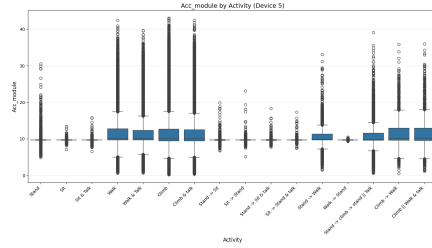
(b) Boxplots of accelerometer by each activity for device 2.



(c) Boxplots of accelerometer by each activity for device 3.

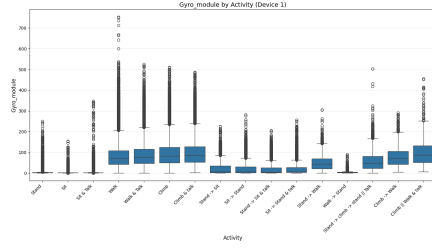


(d) Boxplots of accelerometer by each activity for device 4.

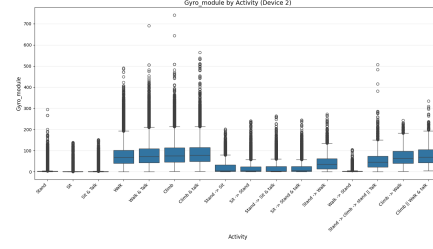


(e) Boxplots of accelerometer by each activity for device 5.

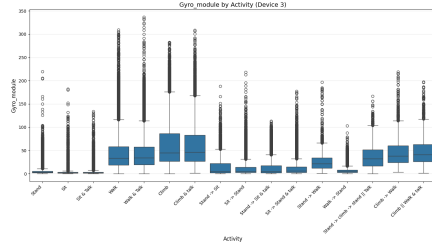
Figure 1: Boxplots of accelerometer by each activity per device.



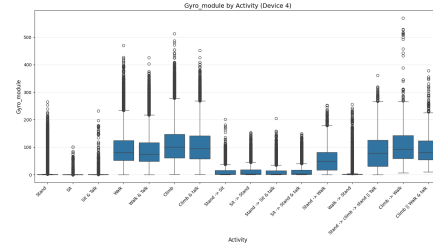
(a) Boxplots of gyroscope by each activity for device 1.



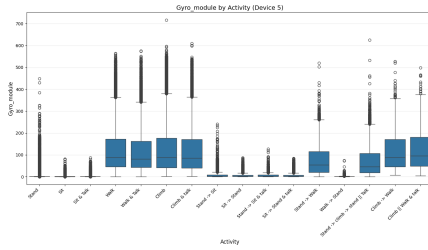
(b) Boxplots of gyroscope by each activity for device 2.



(c) Boxplots of gyroscope by each activity for device 3.

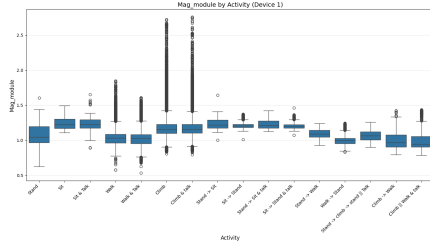


(d) Boxplots of gyroscope by each activity for device 4.

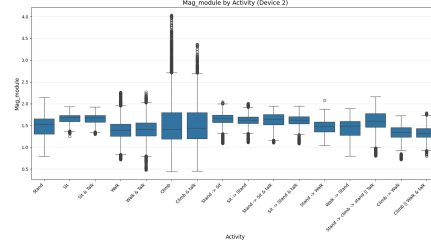


(e) Boxplots of gyroscope by each activity for device 5.

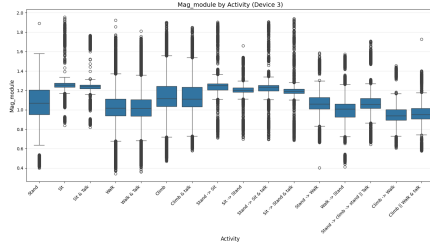
Figure 2: Boxplots of gyroscope by each activity per device.



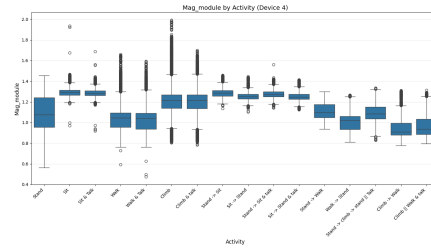
(a) Boxplots of magnetometer by each activity for device 1.



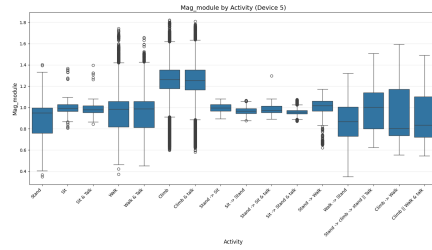
(b) Boxplots of magnetometer by each activity for device 2.



(c) Boxplots of magnetometer by each activity for device 3.

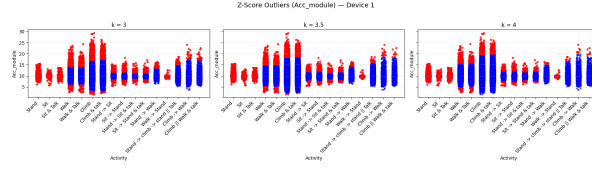


(d) Boxplots of magnetometer by each activity for device 4.

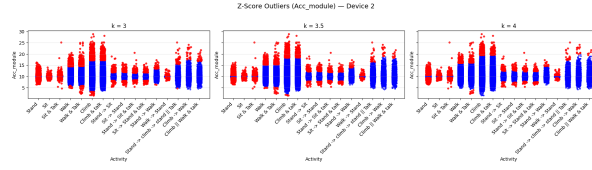


(e) Boxplots of magnetometer by each activity for device 5.

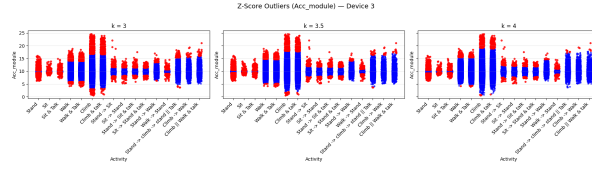
Figure 3: Boxplots of magnetometer by each activity per device.



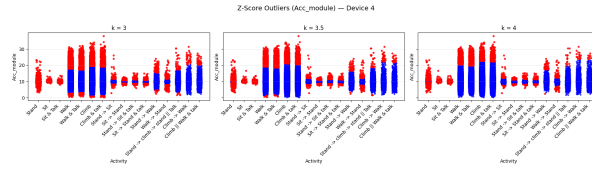
(a) z-score outlier detection of accelerometer by each activity for device 1.



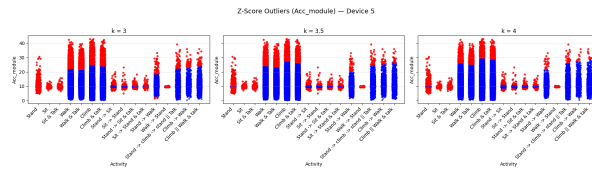
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(c) z-score outlier detection of accelerometer by each activity for device 3.

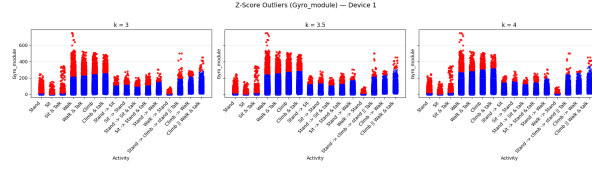


(d) z-score outlier detection of accelerometer by each activity for device 4.

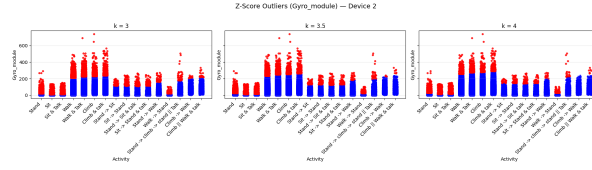


(e) z-score outlier detection of each accelerometer by activity for device 5.

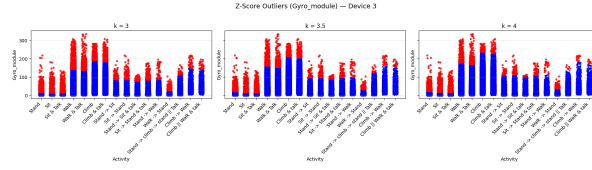
Figure 4: z-score outlier detection of accelerometer by each activity per device.



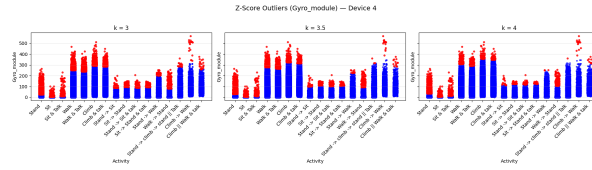
(a) z-score outlier detection of gyroscope by each activity for device 1.



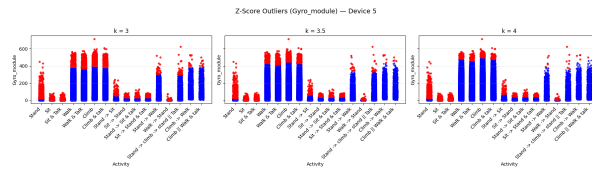
(b) z-score outlier detection of gyroscope by each activity for device 2.



(c) z-score outlier detection of gyroscope by each activity for device 3.

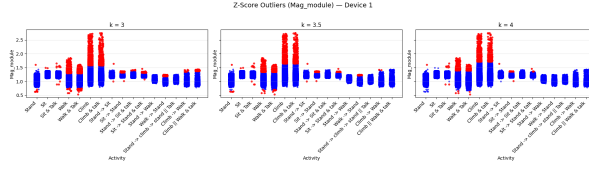


(d) z-score outlier detection of gyroscope by each activity for device 4.

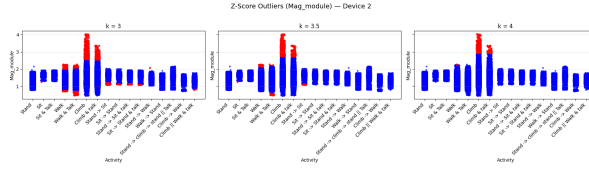


(e) z-score outlier detection of gyroscope each activity for device 5.

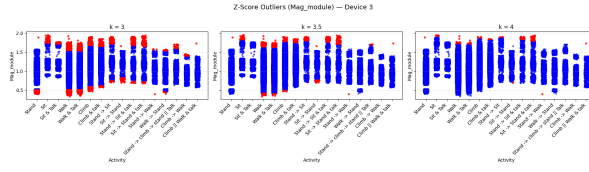
Figure 5: z-score outlier detection of gyroscope by each activity per device.



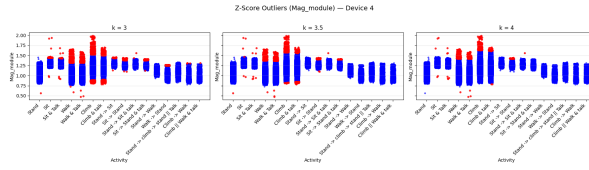
(a) z-score outlier detection of magnetometer by each activity for device 1.



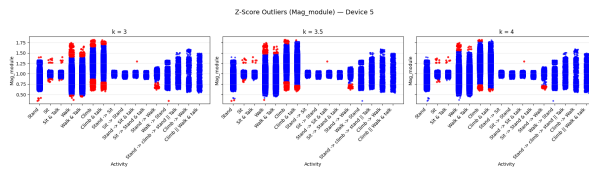
(b) z-score outlier detection of magnetometer by each activity for device 2.



(c) z-score outlier detection of magnetometer by each activity for device 3.

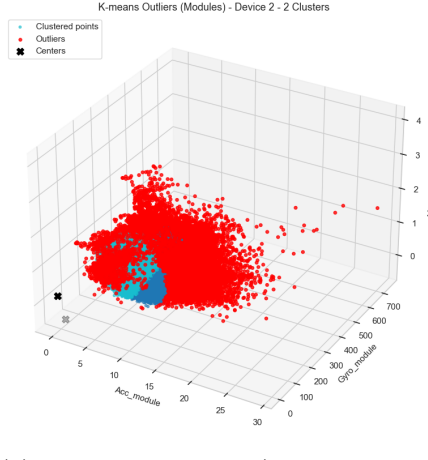


(d) z-score outlier detection of magnetometer by each activity for device 4.

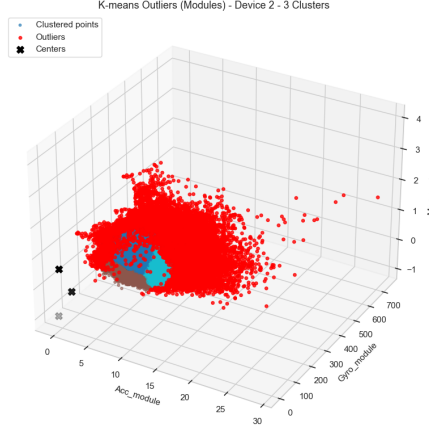


(e) z-score outlier detection of magnetometer each activity for device 5.

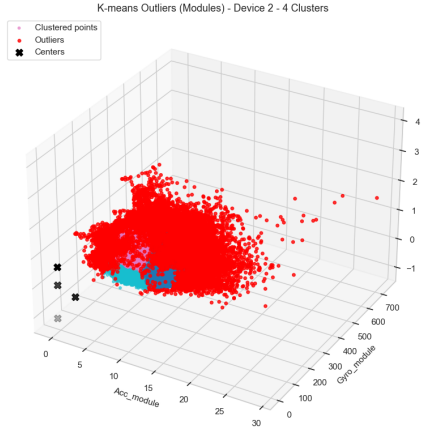
Figure 6: z-score outlier detection of magnetometer by each activity per device.



(a) k-means 3D plot (acc, mag and gyro module) by each activity for device 2 with $k = 2$.



(b) k-means 3D plot (acc, mag and gyro module) by each activity for device 2 with $k = 3$.



(c) k-means 3D plot (acc, mag and gyro module) by each activity for device 2 with $k = 4$.

Figure 7: k-means outlier detection using distanced-based percentil 95 of device 2 with different values of k .

Device 2 | K-Means Outliers | k=3

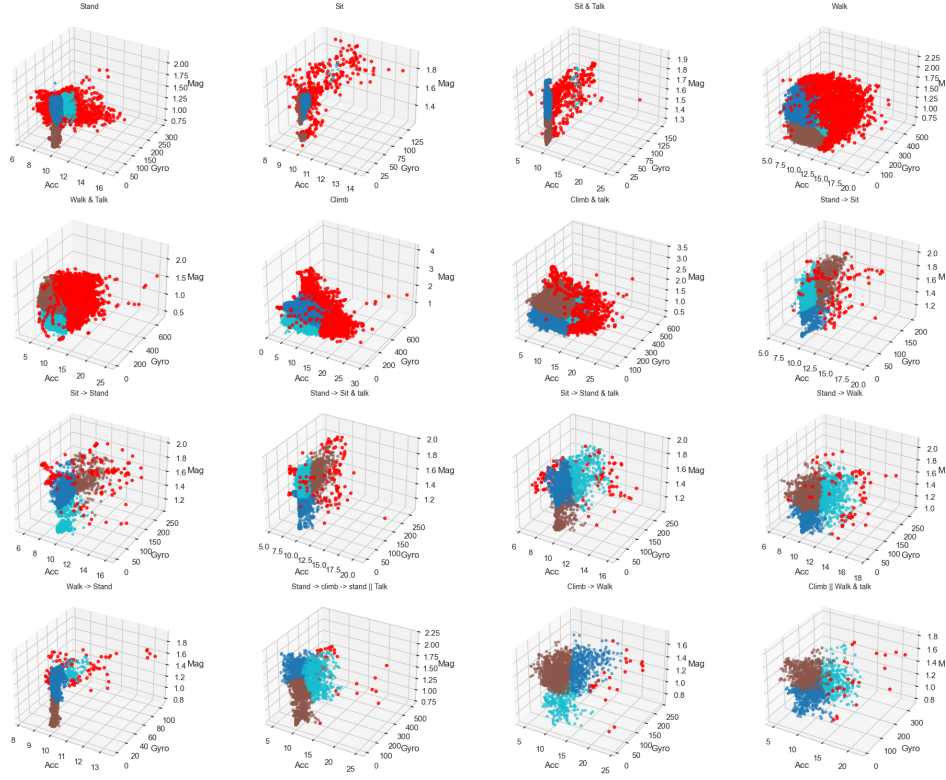


Figure 8: k-means outlier detection using z-score-based normalization for device 2 with $k = 3$.

Device 2 | K-Means Outliers | k=4

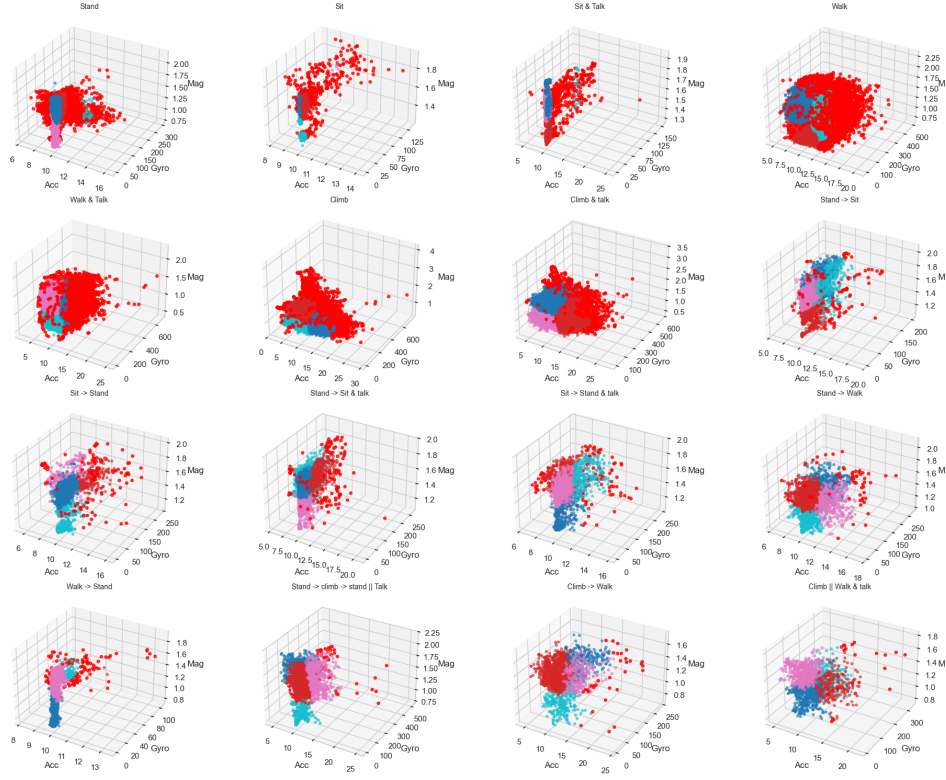


Figure 9: k-means outlier detection using z-score-based normalization for device 2 with $k = 4$.

Device 2 | K-Means Outliers | k=5

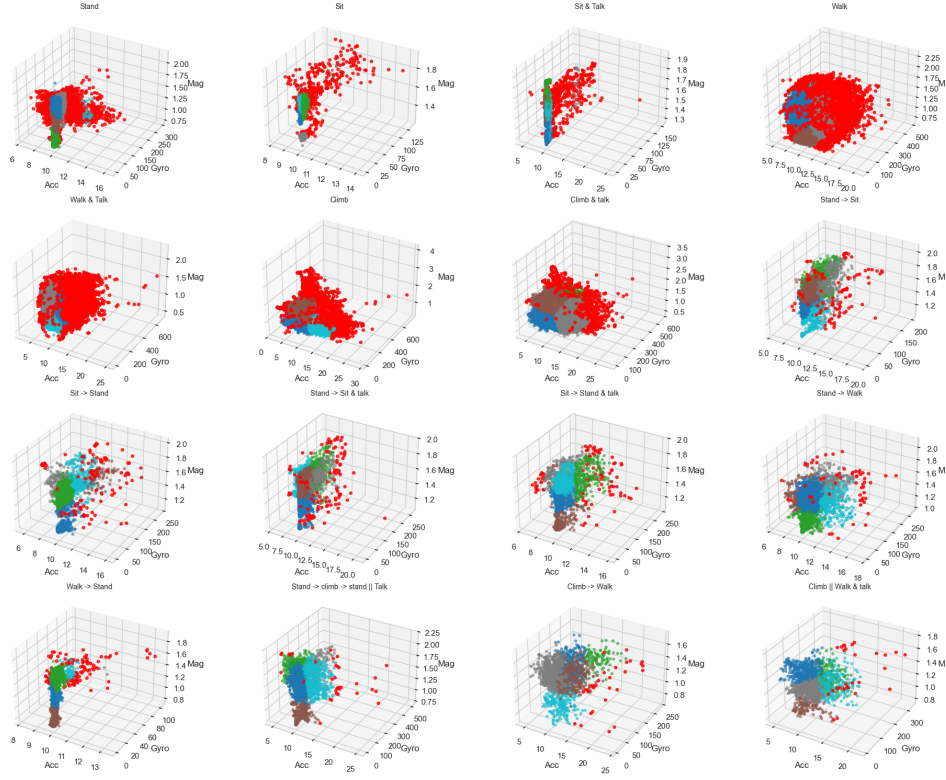


Figure 10: k-means outlier detection using z-score-based normalization for device 2 with $k = 5$.

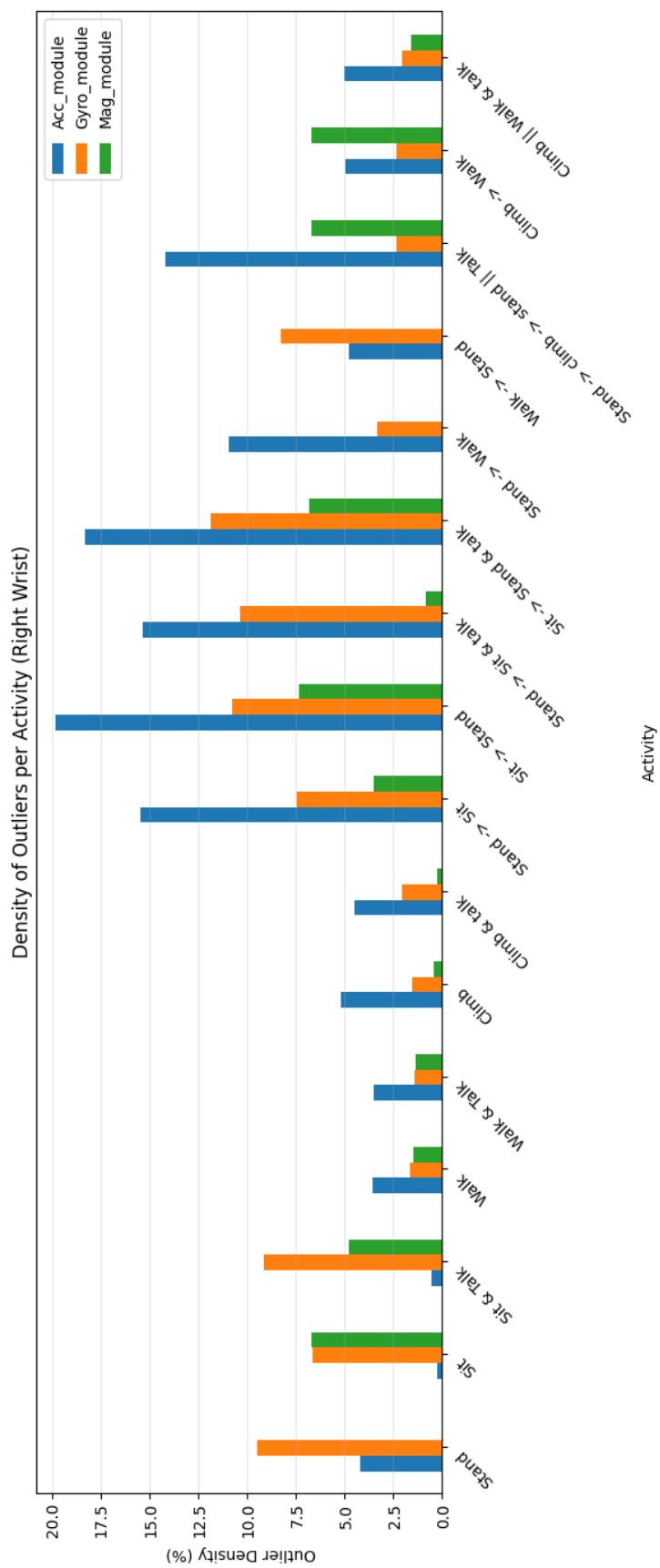


Figure 11: IQR boxplot outlier density for device 2.

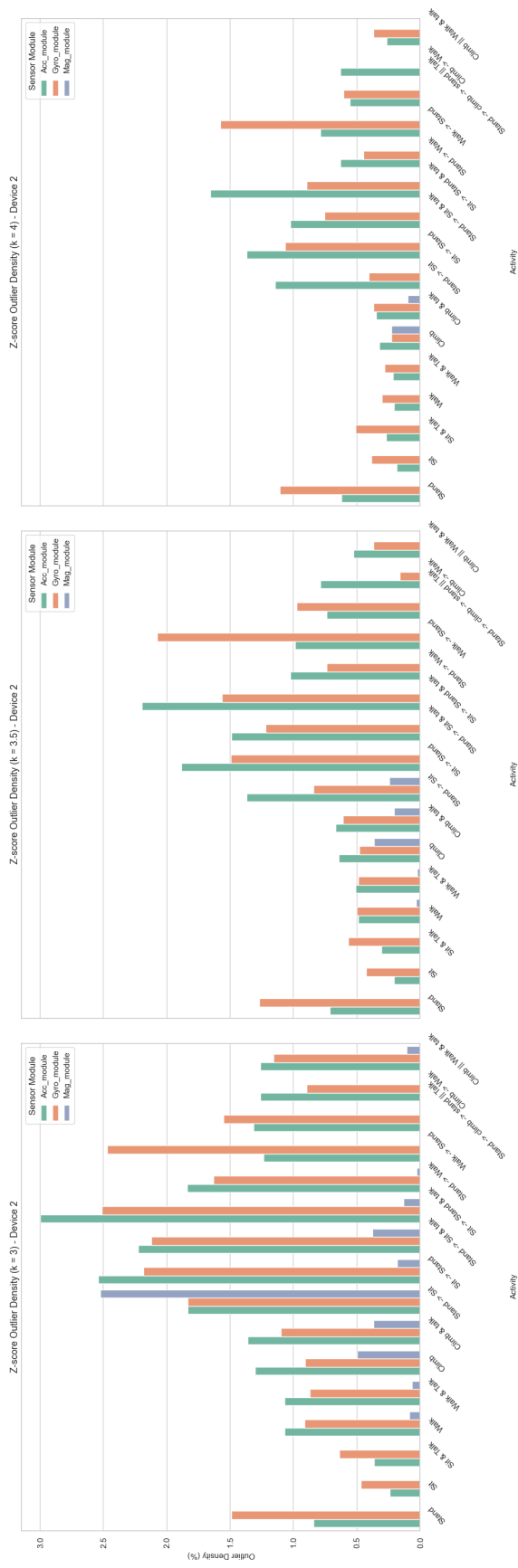


Figure 12: Z-score outlier density for device 2.

K-Means Outlier Density per Activity (Device 2)

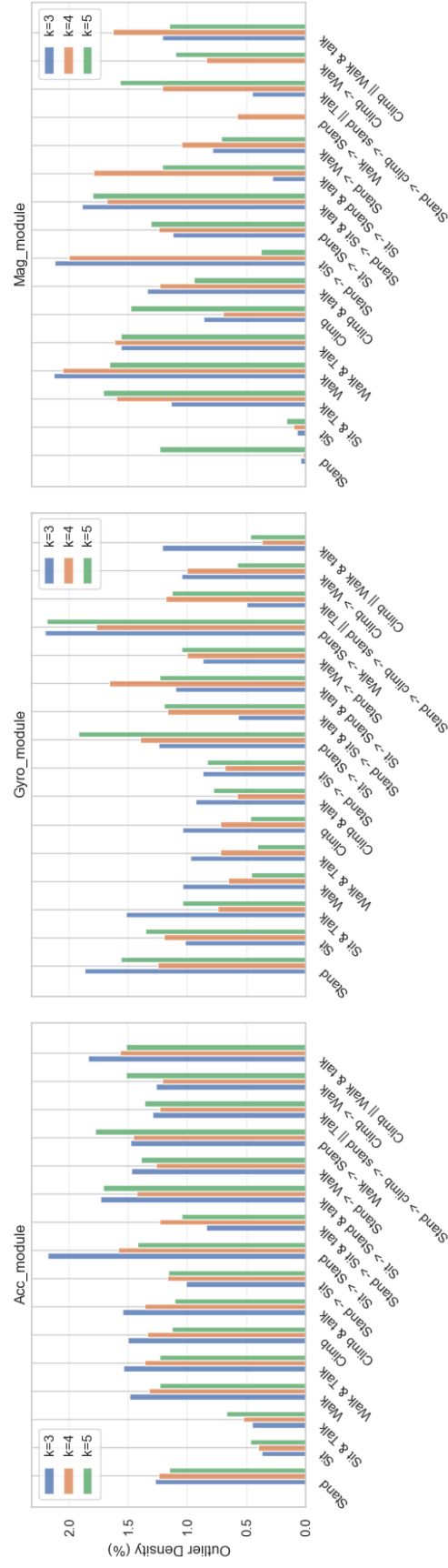


Figure 13: K-means outlier density for device 2.

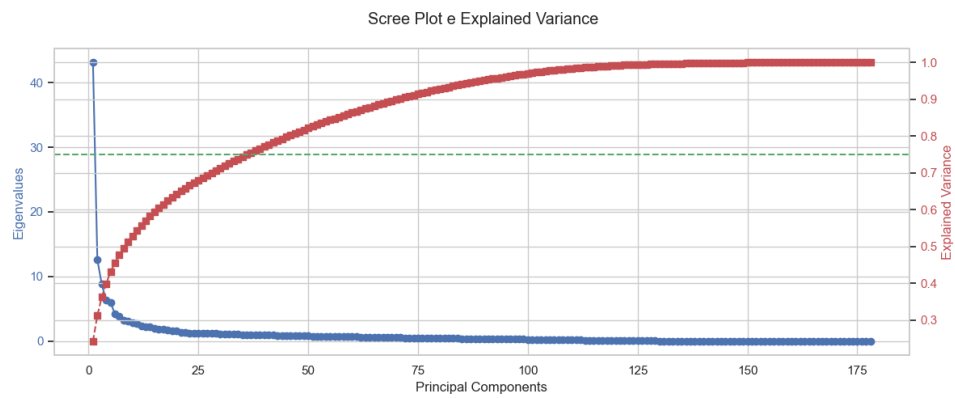


Figure 14: PCA Scree Plot for the new features extracted.

Variable	Normality (KS Test)	Kruskal-Wallis p-value	Significant
Acc_x	False	0.0	True
Acc_y	False	0.0	True
Acc_z	False	0.0	True
Gyro_x	False	0.0	True
Gyro_y	False	0.0	True
Gyro_z	False	0.0	True
Mag_x	False	0.0	True
Mag_y	False	0.0	True
Mag_z	False	0.0	True
Acc_module	False	0.0	True
Gyro_module	False	0.0	True
Mag_module	False	0.0	True

Table 3: Statistical Test Results per Variable

Category	Count
Statistical (per axis)	108
Correlations	36
Physical (core)	8
Physical (per axis)	18
Physical (gravity/gyro)	8
Total	178

Table 4: Final Feature Count by Category

Feature	Fisher Score	Fisher Score (Normalized)
AI	177	1.000000
DF_mx	176	0.994350
gy_mean	175	0.988701
my_skew	174	0.983051
mz_median	173	0.977402
A_bin3	172	0.971752
corr_gy_mx	171	0.966102
EN_max	170	0.960453
ay_skew	169	0.954802
gz_std	168	0.949153

Table 5: Top 10 Fisher Score Features

Feature	ReliefF Score
mx_rms	0.118040
EN_mx	0.102085
mx_median	0.093085
AVH_bin1	0.080090
mz_rms	0.078888
mx_mean	0.078555
corr_mx_mz	0.076095
az_rms	0.076176
corr_gx_gz	0.070697
mz_median	0.070060

Table 6: Top 10 ReliefF Features